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SIMULATION OF FIELD WATER UPTAKE BY PLANTS USING A SOIL WATER DEPENDENT ROOT EXTRACTION FUNCTION

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ABSTRACT

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Water uptake by roots can be represented by adding a volumetric sink term to the continuity equation for soil-water flow. This sink term is often expressed as a product of the difference in pressure head between the soil and the root—soil interface, the hydraulic conductivity of the soil and some empirical root function. Because of the amount of field work and experimental difficulties involved in determining this root function, an attempt was made to describe the sink term with a more simple expression. In this approach the sink term is considered to be a function of the soil-water content, varying with the latter according to the pressure heads generally known to be critical for water uptake by the roots. An implicit finite-difference model was developed and verified with results obtained experimentally in the field from water-balance studies. Although the model does not predict the distribution of soil-water content with depth in very accurate detail, the cumulative effect over the entire depth is properly simulated. Comparison of the results of the simpler model with those of the model of Feddes, Bresler and Neuman, leads to the conclusion that both models yield comparable results.

INTRODUCTION

In describing water uptake by plant roots, one can distinguish between two main approaches. The micro-approach considers the radial soil-water flow to a single root, while the macro-approach takes the integrated properties of the entire root system. In the latter approach water uptake by the roots is represented by a volumetric sink term, which simply is added to the continuity equation for flow of water in soil. In a literature survey, Feddes et al. (1974) show that in most current macroscopic models the sink term is assumed to be directly proportional to the difference in pressure head between the soil and the root interior, to the hydraulic conductivity of the soil and to some empirical root effectiveness function. The interpretation and evaluation of this latter function (which is in effect a coefficient of proportionality of the sink term) is performed differently by various authors. For example, Gardner (1964) and Whistler et al. (1968) take the surface area of the roots per unit volume of soil and some effective distance over which the water moves towards the root — a kind of root-density function. In a similar way, Feddes (1971) considers it as a parameter which accounts for the length and the geometry of the roots. He indirectly derived this parameter from vertical flow experimentally obtained from field lysimeter and γ-transmission data, assuming that the rate of vertical flow through the soil—plant—atmosphere system is linearly proportional to the corresponding pressure differences between these media.

Nimah and Hanks (1973) consider the root-distribution function in a depth interval as the weight fraction of the roots relative to the total weight of the roots. They determine this function by sampling the soil profile in the field.

Feddes et al. (1974) showed that the root effectiveness function is proportional to root mass and that both varied nearly exponentially with depth. They suggest that this function can be found by sampling the root mass with depth and determining the proportionality constant from model calibration.

It is clear that the root effectiveness function under field conditions will vary with the type of soil and rooting system. Moreover, the latter usually changes with depth and time. Therefore, the determination of the effectiveness function may take a lot of work or may need careful and expensive experimentation. For this reason, a different approach is proposed in which the water uptake by roots is considered to be a function of the water content of the soil.

THEORY

When describing the flow of water in unsaturated—saturated soil systems it is customary to use Darcy's law. For one-dimensional vertical flow, with the origin of z (cm) at the soil surface and positive in downward direction, the volumetric flux q (cm^3 $H_2O \cdot cm^{-2}$ soil $\cdot sec^{-1}$) can be written as:

$$q = -K(\theta)\frac{\partial H}{\partial z} \qquad (1)$$

where θ is the volumetric water content of the soil (cm^3 $H_2O \cdot cm^{-3}$ soil), $K(\theta)$ is the hydraulic conductivity of the soil ($cm \cdot sec^{-1}$), H is the total head (cm), i.e., the sum of the pressure head $h(\theta)$ and the gravitational head $-z$, $H = h - z$.

Applying the principle of continuity and representing the water uptake by the roots as a sink term S (cm^3 $H_2O \cdot cm^{-3}$ soil$\cdot sec^{-1}$) depending on θ, the time rate of change of the soil-water content is:

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z} - S(\theta) \qquad (2)$$

where t is the time (sec).

Combining eq.1 with eq.2 gives the basic differential equation for one-dimensional vertical soil-water flow:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(K(\theta) \frac{\partial H}{\partial z} \right) - S(\theta) \tag{3}$$

Eq.3 is a second-order, parabolic partial-differential equation which is non-linear because of the dependence of K upon θ. Substituting $H = h - z$, assuming a single-valued relationship between h and θ (i.e., no hysteresis taken into account) and representing $K \partial h / \partial \theta$ by the symbol of the diffusivity D ($cm^2 \cdot sec^{-1}$), eq.3 becomes:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(D(\theta) \frac{\partial \theta}{\partial z} \right) - \frac{\partial K(\theta)}{\partial z} - S(\theta) \tag{4}$$

In order to be able to solve eq.4, one has to define the sink term. For the functional dependence of the sink term upon the soil-water content some assumptions can be made (see Fig.1).

For conditions drier than "wilting point" and wetter than a certain "anaerobiosis point", it may be assumed that the water uptake by the roots is zero. It is known that there exists no single "wilting point" because it varies with the variation in osmotic pressure of plant-tissues. The variation in soil-water pressure head for wilting varies from – 10,000 to – 20,000 cm, but for many practical purposes a mean value of about – 15,000 cm can be regarded as an important soil value.

A fixed "anaerobiosis point" at which deficient aeration conditions exist and root growth is seriously hampered, is hard to define. To characterize soil aeration, one can use the so-called oxygen diffusion rate (ODR), which is defined as the flux of oxygen towards a platinum wire inserted in the soil. Stolzy

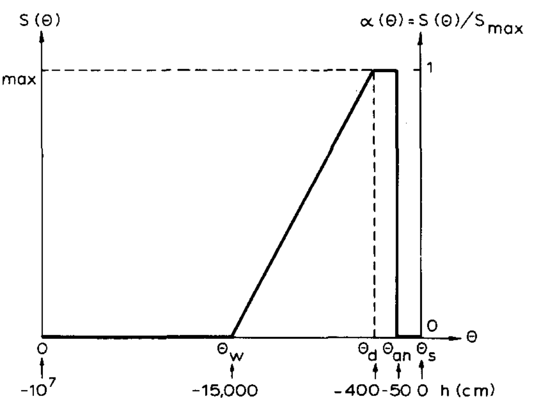

Fig.1. General shape of the sink term as a function of soil-water content. Also indicated are the pressure heads assumed to be critical for water uptake by the roots as explained in the text.

and Letey (cf. Wesseling, 1974) found that many plants do not grow in soils with oxygen diffusion rates (ODR) below $20 \cdot 10^{-8}\,g \cdot cm^{-2} \cdot min^{-1}$. For a sandy loam of single-grained structure with a pore volume of $0.50\,cm^3 \cdot cm^{-3}$ this rate is found at a soil-water pressure head of – 50 cm and a gas-filled porosity of $0.10\,cm^3 \cdot cm^{-3}$ (Bakker, 1970). However, for different soils different relationships between ODR and soil-water pressure heads are found (Kowalik, 1972).

Soil aeration can also be characterized by the possible gas exchange through the gas-filled soil pores, i.e., by the gas diffusion coefficient. From calculations on the necessary transport of oxygen towards the roots of normal growing plants, it appears that below an oxygen diffusion coefficient of $1.5 \cdot 10^{-4}\,cm^2 \cdot sec^{-1}$ the oxygen demand of the plants can never be met (Bakker et al., 1976). For single-grained structures, this value is found at a gas-filled porosity of $0.10\,cm^3 \cdot cm^{-3}$, for good structured soils, however, at a gas porosity of $0.04\,cm^3 \cdot cm^{-3}$. In the case of the clay soil here investigated, an air content of 0.04 corresponds to a pressure head of about – 50 cm. Therefore we assumed that no water uptake by the roots takes place above the anaerobiosis point which corresponds in the two cases mentioned to a pressure head of – 50 cm.

From a review on moisture requirements and effects of pressure head on yield and quality of various vegetable crops, Feddes (1971) concluded that in general for these crops the pressure head at which soil water begins to limit plant growth is about – 400 cm. Now we assume that the water uptake by the roots is constant and at a maximum rate for $-400 < h < -50$ cm. The water uptake for the plant, S, is assumed to decrease linearly with θ between $h = -400$ cm and $h = -15{,}000$ cm.

Fig.1 shows S as a function of θ over the whole soil water content range. Also indicated are the corresponding soil water pressure heads.

For convenience sake, the dimensionless variable:

$$\alpha(\theta) = S(\theta)/S_{max} \tag{5}$$

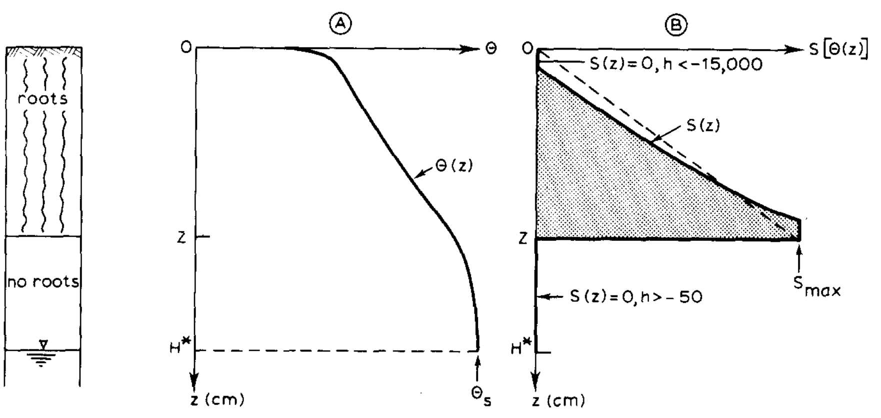

Fig.2. General patterns for the variation of soil-water content (A) and that of the sink term (B) with depth.

is introduced. This variable therefore runs from 0 to 1, as is shown in Fig.1.

In Fig.2 an example of the variation in soil-water content θ and in sink term S with depth z is given. The actual transpiration E_{pl} ($cm \cdot sec^{-1}$) is:

$$E_{pl} = \int_0^Z S(z,t)\,dz \tag{6}$$

where Z is the rooting depth (cm). Since the upper dry part in Fig.2B generally is small, the integral of eq.6 can be approximated by the area of the triangle $0ZS_{max}$ and in this case we have:

$$E_{pl} = \tfrac{1}{2} \cdot Z \cdot S_{max}$$

so:

$$S_{max} = \frac{2E_{pl}}{Z} \tag{7}$$

The example in Fig.2 holds for a soil with a shallow groundwater table. For soils without the presence of a groundwater table, the variation of θ with z will be different, depending on the antecedent conditions of rainfall or irrigation, of evapotranspiration, etc. In that case the variation of S with z will be different, and therefore the shape of the triangle will change. However, the area of the triangle will always remain the same and can still be represented by eq.7.

Since from eq.5 $S(\theta) = \alpha(\theta) S_{max}$, the expression for the sink term becomes:

$$S(\theta) = \alpha(\theta)\,\frac{2E_{pl}}{Z} \tag{8}$$

Substitution of eq.8 into eq.4 yields:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left(D(\theta)\frac{\partial \theta}{\partial z}\right) - \frac{\partial K(\theta)}{\partial z} - \alpha(\theta)\,\frac{2E_{pl}}{Z} \tag{9}$$

NUMERICAL APPROACH

To solve eq.9a numerically, one can set the following initial and boundary conditions:

$$\theta = \theta(z,0), \qquad t = 0, \qquad 0 \leqslant z \leqslant H^{\star} \tag{10a}$$
$$\theta = \theta(0,t), \qquad t > 0, \qquad z = 0 \tag{10b}$$
$$\theta = \theta_s(H^{\star},t), \qquad t > 0, \qquad z = H^{\star} \tag{10c}$$

where $H^{\star}$ is the depth of the groundwater table and the subscript s denotes saturated condition. The soil-water content at the surface and the groundwater table depth must be known as functions of time.

If the relative humidity (f) and the temperature of the air (T) as a function of time are known, and if it may be assumed that the pressure head at the soil surface is at equilibrium with the atmosphere, then $h(0,t)$ can be derived from the well-known relationship:

$$h(0,t) = \frac{RT(t)}{Mg} \ln [f(t)] \tag{11}$$

where R is the universal gas constant ($\mathrm{erg \cdot mole^{-1} \cdot K^{-1}}$), T is the absolute temperature (K), g is the acceleration of gravity ($\mathrm{cm \cdot sec^{-2}}$), M is the molecular weight of water ($\mathrm{g \cdot mole^{-1}}$) and f is the relative humidity of the air. Knowing $h(0,t)$, $\theta(0,t)$ can be derived from the soil water retention curve. As water losses from soil and plant cannot exceed the demand of the atmosphere, i.e., cannot exceed potential evapotranspiration, upper limits for the actual soil flux q and integrated plant root extraction rate $E_{\mathrm{pl}}(t)$ are to be known. Therefore the maximum possible flux at the soil surface $|\chi(t)|$ and the maximum possible flux through the plant leaf surface $E^{\star}_{\mathrm{pl}}(t)$ must be given functions of time.

In calculating values of maximum possible evapotranspiration for a given crop one may use the combination type equation (e.g., Feddes, 1971):

$$E^{\star}(t) = \frac{(R_{\mathrm{n}} - G)/L + \rho_{\mathrm{a}} c_{\mathrm{p}} (\epsilon^{\star}_{\mathrm{z}} - \epsilon_{\mathrm{z}})/r_{\mathrm{a}}}{(\delta + \gamma)} \tag{12}$$

where R_{n} is net radiation flux ($\mathrm{W \cdot m^{-2}}$); G is heat flux into the soil ($\mathrm{W \cdot m^{-2}}$), L is latent heat of vaporization of water ($\mathrm{J \cdot kg^{-1}}$), ρ_{a} is density of moist air ($\mathrm{kg \cdot m^{-3}}$) c_{p} is specific heat of air at constant pressure ($\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$), δ is slope of the saturation vapor pressure curve ($\mathrm{bar \cdot K^{-1}}$), $\epsilon_{\mathrm{z}}, \epsilon^{\star}_{\mathrm{z}}$ is unsaturated and saturated water-vapor pressure in the air at height z and temperature T_{z} (bar), r_{a} is diffusion resistance to water vapor of the air layer surrounding the leaves ($\mathrm{sec \cdot m^{-1}}$), γ is psychometric constant ($\mathrm{bar \cdot K^{-1}}$). Values of r_{a} for various crop heights and wind velocities can be found elsewhere (e.g., Feddes, 1971, table 6).

To calculate the maximum possible rate of evaporation from the soil, one can use a simplified form of eq.12 which neglects the aerodynamic term and takes into account the fraction of R_{n} received at the soil surface (Ritchie, 1972):

$$E^{\star}_{\mathrm{s}}(t) = \chi(t) = \frac{\delta}{(\delta + \gamma)L} R_{\mathrm{n}} \mathrm{e}^{-0.39\ \mathrm{LAI}} \tag{13}$$

where LAI is a so-called leaf area index (note that under non-evaporative conditions, $\chi(t)$ is equal to the measured rainfall). By subtracting $E^{\star}_{\mathrm{s}}(t)$ from $E^{\star}(t)$ one obtains the maximum possible transpiration rate $E^{\star}_{\mathrm{pl}}(t)$. For the present study mean daily values (24 h) of meteorological data were used to determine $E^{\star}(t)$ and $E^{\star}_{\mathrm{s}}(t)$.

In the numerical approach, eq.4 is approximated by the following finite-difference solution. Let a grid be drawn over a scale diagram of profile depth and time and let the index along the space ordinate be denoted by i and the index along the time abscissa by j. Then an implicit formulation for eq.4 is:

$$\frac{\theta_{i,j+1}-\theta_{i,j}}{\Delta t_j} \approx \frac{1}{\Delta z}\left\{\left[D(\theta_{i+½,j+½})\left(\frac{\partial\theta}{\partial z}\right)_{i+½,j+½} - K(\theta_{i+½,j+½})\right.\right.$$

$$- \Delta z ½ S(\theta_{i+½,j+½}) \quad - \quad D(\theta_{i-½,j+½})\left(\frac{\partial\theta}{\partial z}\right)_{i-½,j+½}$$

$$\left.\left. - K(\theta_{i-½,j+½}) + \Delta z ½ S(\theta_{i-½,j+½})\right]\right\} \tag{14}$$

where:

$$\left(\frac{\partial\theta}{\partial z}\right)_{i+½,j+½} \approx \frac{1}{2\Delta z}\left[(\theta_{i+1,j+1}+\theta_{i+1,j})-(\theta_{i,j+1}+\theta_{i,j})\right] \tag{15a}$$

$$\left(\frac{\partial\theta}{\partial z}\right)_{i-½,j+½} \approx \frac{1}{2\Delta z}\left[(\theta_{i,j+1}+\theta_{i,j})-(\theta_{i-1,j+1}+\theta_{i-1,j})\right] \tag{15b}$$

$$\theta_{i+½,j+½} \approx \frac{1}{2} \quad 1+\frac{\Delta t_j}{2\Delta t} \quad (\theta_{i+1,j}+\theta_{i,j})$$

$$-\frac{1}{4}\frac{\Delta t_j}{\Delta t_{j-1}}(\theta_{i+1,j-1}+\theta_{i,j-1}) \tag{15c}$$

$$\theta_{i-½,j+½} \approx \frac{1}{2} \quad 1+\frac{\Delta t_j}{2\Delta t_{j-1}} \quad (\theta_{i,j}+\theta_{i-1,j})$$

$$-\frac{1}{4}\frac{\Delta t_j}{\Delta t_{j-1}}(\theta_{i,j-1}+\theta_{i-1,j-1}) \tag{15d}$$

When applied at all nodes of the depth—time diagram, the result is a system of simultaneous linear algebraic equations with a tridiagonal coefficient matrix and unknowns at the $j+1$ time level. With boundary conditions specified, the number of equations equals the number of unknowns, and the system of equations can be solved by the tridiagonal matrix algorithm. For calculation examples with such a model, see Kowalik and Zaradny (1974).

The input into the model consists of $h(\theta)$, $K(\theta)$ and $\alpha(\theta)$ relationships as well as the values of $\theta_0(z)$, $\theta(0,t)$, $\theta_s(H^\star,t)$ and $L(t)$. Also $\chi(t)$ and $E^\star_{pl}(t)$ must be specified at each instant. The output data are $\theta(z,t)$, $K(z,t)$, $h(z,t)$, $q(0,t)$, $q(H^\star,t)$, $S(z,t)$ and cumulative $E_{pl}(t)$.

FIELD EXPERIMENT

A field experiment was performed by the first author in 1967 (Feddes, 1971) at the groundwater-level experimental field Geestmerambacht in The Netherlands. Red cabbage (*Brassica oleracea* L. c.v. "Rode Herfst") was grown on a heavy clay soil (48% < 2 μm) in rows 75 cm wide and spaced 65 cm apart.

Water-balance studies were carried out by means of a non-weighable lysimeter in which the groundwater table was maintained within 1 cm at the same level as that in the surrounding field. Upward and downward flows from and to the water table were measured daily in the lysimeter. Soil-water content measurements were performed weekly at 10-cm depth intervals with the aid of the γ-transmission method. Precipitation was measured at a neighboring weather station with a recording rain gauge having its rim level at ground surface. The actual evapotranspiration was the only unknown in the water-balance equation and could therefore be easily calculated for each week.

Soil-water retention curves (drying) were determined in the laboratory without considering hysteresis. Hydraulic conductivity data were obtained in the laboratory by an infiltration method (Wesseling and Wit, 1966) and in the field from flow measurements during dry periods (Feddes, 1971). All meteorological data needed for the calculation of the surface water content (eq.11) of maximum possible transpiration as well as soil evaporation (see eqs.12 and 13) were derived from measurements of air temperature, relative humidity, wind velocity, and solar radiation at the weather station, and from net radiation above the cabbage crop.

Growth and development of the crop were analyzed at regular time intervals by measuring crop height, soil-cover fraction, leaf area and rooting depth.

RESULTS AND DISCUSSION

To apply the finite difference method, the soil was divided into layers of 5 cm, starting at $z = 2.5$ cm. For the time step a fixed value of 2 h was taken. The bulk density of the soil varied with depth, i.e., increased from values of about 0.90 $g \cdot cm^{-3}$ at the surface to about 1.35 $g \cdot cm^{-3}$ at depths of 25—35 cm and remained more or less constant below these depths (Feddes, 1971). This should be reflected in the properties assigned to each depth interval. However, as previously done by Feddes et al. (1974), the soil profile was considered to be homogeneous and a single water retention curve was adopted, which corresponds to a bulk density of 1.34 $g \cdot cm^{-3}$ (Fig.3). The hydraulic conductivity data from various depths, as determined in the field and in the laboratory, were similarly lumped into a single $K(\theta)$-relationship representing the entire soil profile (Fig.4).

The dimensionless sink function $\alpha(\theta)$ can be constructed according to the critical h values discussed earlier and presented in Fig.1. These h values are: -10^7, $-15{,}000$, -400, -50 and 0 cm with corresponding θ values (Fig.4) of 0, 0.25, 0.435, 0.47 and 0.50, respectively. From Fig.1 it follows then that $\alpha(\theta) = 0$ for $0 \leqslant \theta \leqslant 0.25$ and for $0.47 < \theta \leqslant 0.50$, $\alpha(\theta) = 1$ for $0.435 < \theta < 0.47$, and $\alpha(\theta)$ is linearly increasing from 0 to 1 for $0.25 < \theta \leqslant 0.435$. This leads to the values given in Table I.

In the numerical approach, the growth of the roots is taken into account by

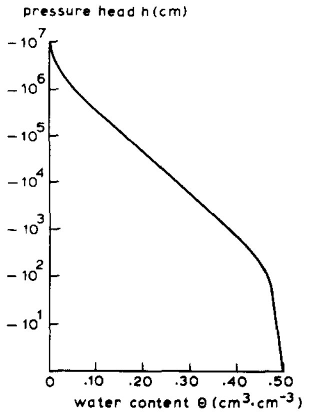

Fig.3. Relation between pressure head and volumetric soil-water content for a clay at a dry bulk density of 1.34 g·cm^{-3}.

using different values of the rooting depth Z at different times. The depth of the effective root zone varied from about 25 cm at the beginning of the 7-week-experiment to about 70 cm at the end of it.

Fig.5 shows the variation of the soil-water content at the surface with time

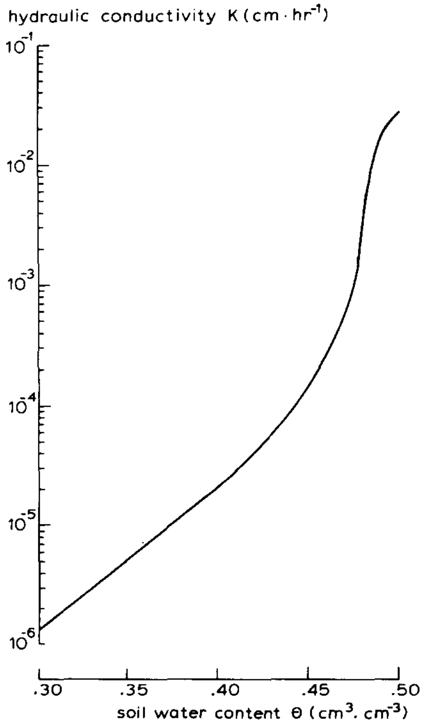

Fig.4. Relation between hydraulic conductivity and soil-water content for a clay.

TABLE I

Dimensionless sink term (* obtained by interpolation) as a function of soil-water content for clay soil

θ ($cm^3 \cdot cm^{-3}$)	$\alpha(\theta) = S(\theta)/S_{max}$	θ ($cm^3 \cdot cm^{-3}$)	$\alpha(\theta) = S(\theta)/S_{max}$
⩽ 0.25	0	0.43	0.974*
0.26	0.054*	0.435	1.000
0.30	0.270*	0.47	1.000
0.35	0.540*	>0.47	0
0.40	0.811*		

as derived by means of eq.11 and Fig.3. During periods of rain θ (0) was calculated by distributing the rainfall over the 2-h period over the top layer of 5 cm. Thus over this depth range θ was considered to be constant for each period. For details on the variation of maximum possible transpiration and soil evaporation the reader is referred to Fig.6.

A comparison between measured and calculated soil-water content profiles for the period from June 27 through August 9 is given in Fig.7. The experimental data represent average values obtained from two to three repeated measurements with the γ-transmission method. The agreement for the top 20 cm is not very good during the first 3 weeks, but quite good during the last 4 weeks. The agreement within the depth interval between 30 and 70 cm is satisfactory during the first 4 weeks. However, during the last 3 weeks the computations yield drier conditions than the measured ones. As far as the depth interval between 70 and 100 cm is concerned we may conclude that there is satisfactory agreement throughout the course of the experiment.

Discrepancies between measured and computed data may be due to the adoption of a single moisture-retention curve and a single hydraulic-conductivity curve for the entire soil profile. The importance of the $K(\theta)$

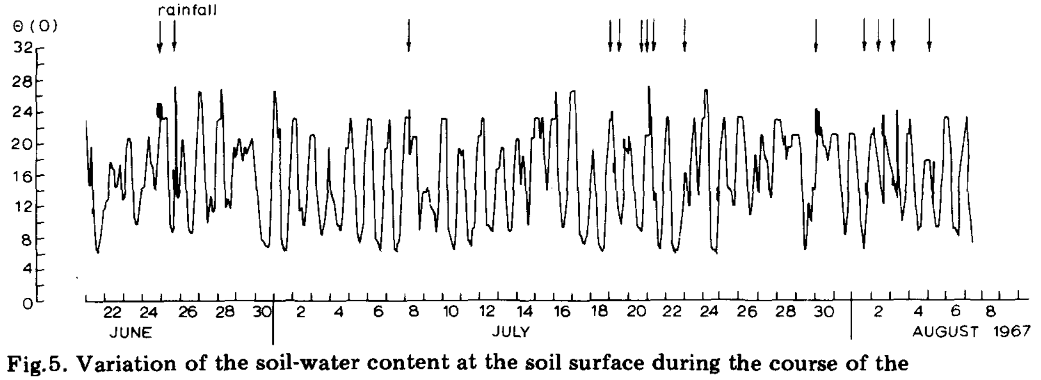

Fig.5. Variation of the soil-water content at the soil surface during the course of the experiment.

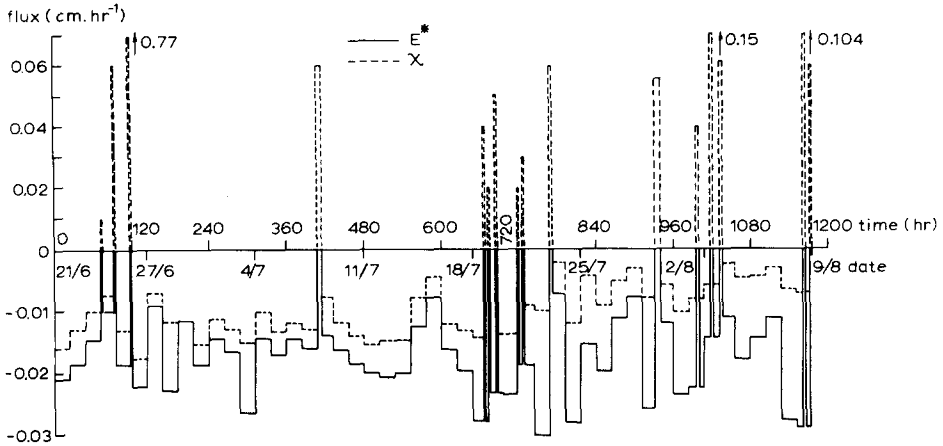

Fig.6. Variation with time of calculated maximum possible evapotranspiration E^* (solid line) and calculated maximum possible flux through the soil surface χ (dashed line). The χ is measured rain when it is positive and is calculated maximum soil evaporation E_s^* when it is negative (after Feddes et al., 1974).

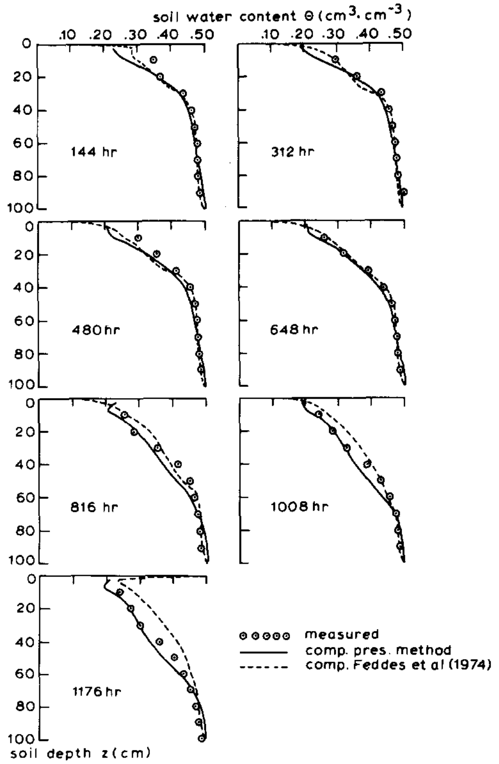

Fig.7. Comparison between measured and computed soil-water content profiles at various numbers of hours after the start of the experiment.

relationship is further revealed if one compares the cumulative upward flow as computed with that measured by the lysimeter (Fig.8). The agreement is close during the first half of the period of observation. During the second half, measurement and computation diverge.

In Fig.7 the results obtained by Feddes et al. (1974) with a principally different extraction term for the description of the water uptake by the roots are also given. During the first 4 weeks the results of both approaches are similar. During the last 3 weeks the present results are too dry in the middle portion of the profile, whereas those of Feddes et al. (1974) are too wet in the upper portion of the profile. The differences between the results of the two methods may be due to the different way in which the boundary condition at the top was handled and to the difference in description of the water uptake by the roots. Both methods show that the discrepancies largely occur in the root zone, which may stem from an improper distribution of the extraction term throughout the root zone by the oversimplified expressions used for the sink term S in both approaches. Fig.9 shows how the sink term changes as a function of time and depth. Since the actual sink term could be derived only from lysimeter and soil-water data as an average over periods of one week, the calculated values cannot be compared directly with results of field data. The shape and order of magnitude, however, are similar to the time-averaged root extraction rates derived from field data of the same profile (Feddes, 1971). From Fig.9 it is seen that the magnitude of the sink is small in the upper layers of the profile, increases to a certain maximum and decreases to zero at the bottom of the root zone. The zone of maximum sink activity moves downward with time and water uptake from the upper layers becomes relatively less important. Similar results were reported by Reicosky et al. (1972) and Arya et al. (1975) on soybeans, by Stone et al. (1973) on sorghum and by Rice (1975) on bermudagrass. The maximum appears to depend on the demand the atmosphere sets on the plant system, on the depth to which the roots penetrate, and on the soil water content.

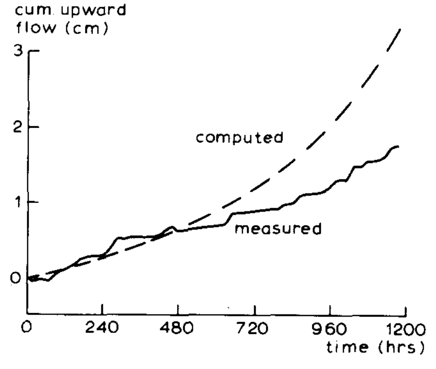

Fig.8. Comparison between computed and measured cumulative upward flow during the period considered.

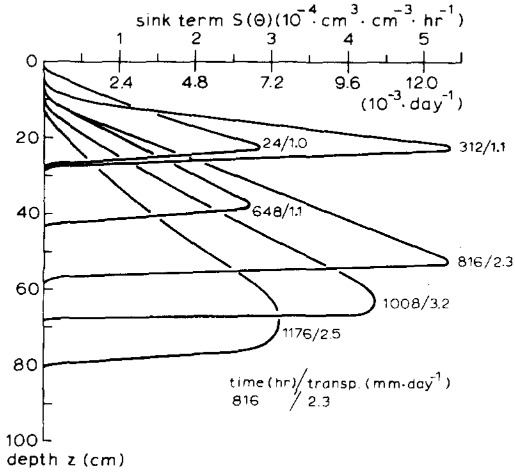

Fig.9. Computed root extraction rates as a function of depth at various times of the 49-day period considered.

Although the present model and that of Feddes et al. (1974) do not give accurate data on soil-water profiles, the cumulative water uptake over the entire depth of the root zone is properly simulated with both methods. In Fig.10 the cumulative amounts of water leaving a unit horizontal soil surface area by plant transpiration and soil evaporation are shown as functions of time. There is an excellent agreement between the transpiration and soil-evaporation data measured in the field by the lysimeter and those computed by the present model. The results of the computations of Feddes et al. (1974) yield somewhat higher transpiration values during the first three weeks, during the last

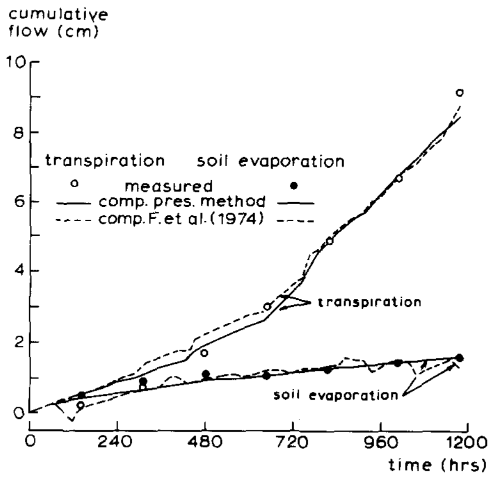

Fig.10. Comparison of experimental and computed soil evaporation and plant transpiration for red cabbage on a clay soil.

4 weeks they are the same, however. Part of this discrepancy may stem from the difference in applied top boundary condition and extraction term.

The soil evaporation computed by Feddes et al. (1974) shows somewhat more variation with time than does the present model. This may also be due to the differences in top boundary conditions applied. However, the final volume computed by the two methods is practically the same.

From the above it can be concluded that, at least in this case, the present method, which requires less experimental work, yields results as good as those of the other model referred to.

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